

GPU Device Information Report

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1 Objective

The purpose of this labwork was to use `numba.cuda` to select a CUDA device and display basic information about the GPU.

2 Results

The selected device was device 0. The information obtained from the program is summarized below.

Property	Value
Device ID	0
Device name	NVIDIA GeForce RTX 3060
Multiprocessor count	28
Cores per multiprocessor	128
CUDA core count	3584
Compute capability	8.6
Total memory	12.0 GiB (12,884,377,600 bytes)
Free memory at execution	10.98 GiB (11,789,139,968 bytes)

There is no built-in function in `numba.cuda` to directly retrieve the number of CUDA cores. However, the number of CUDA cores can be calculated using the formula:

$$\text{CUDA cores} = \text{Multiprocessor count} \times \text{Cores per multiprocessor}$$

3 Captured Output

Figure 1 shows the output produced by the Python program.

```
PS D:\000Thac Si\advancedhpc2026> python -u "d:\000Thac Si\advancedhpc2026\Labwork 2\check_my_gpu.py"
Device Name: b'NVIDIA GeForce RTX 3060'
Device ID: 0
Device Multiprocessor Count: 28
Device Compute Capability: (8, 6)
Device memory size: MemoryInfo(free=11789139968, total=12884377600)
```

Figure 1: GPU information displayed by the `numba.cuda` program.

4 Conclusion

The `numba.cuda` library successfully detected and selected the NVIDIA GPU. It provided the device name, ID, multiprocessor count, compute capability, and memory information.